THE MALAWI NATIONAL EXAMINATIONS BOARD

MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

AGRICULTURE

Subject Number: M012/II

Time Allowed: $1\frac{1}{2}$ hour sessions

SAMPLE PAPER

PAPER II

(40 marks)

Practical

Instructions

1. **This paper contains 8 printed pages. Please check.**
2. This paper contains two sections **A** and **B**. Section A has two descriptive questions and Section B has two questions on experiment.
3. Answer **all** questions in the spaces provided. Marks are indicated against each part of the question.
4. Write your **Examination Number** on **all** pages.
5. In the table provided on this page, **tick** against the question number you have answered.
6. At the end of the examination, hand in your paper to the invigilator.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			

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SECTION A (20 MARKS)

1. A teacher gave students the following items:

- 50 g of dry soil.
- Sieves of the following mesh sizes 0.2 mm, 0.02 mm and 0.002 mm.
- A weighing scale.

Describe an experiment that would be done to determine percentage of different soil particles of the sample. **(10 marks)**

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1. (Continued)

2. **Table below** shows service life in years and value in Kwacha of a spray race. Use it to answer questions that follow:

Service Life in Years	Value (MK)
1	150,000
2	130,000
3	110,000
4	90,000

a. Plot a graph showing value of the spray race on **Y**-axis and service life in years on **X**-axis. **(3 marks)**

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2. a. **(Continued)**

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2. **(Continued)**

b. From the graph plotted in **2 a** state the relationship between value of the spray race and its service life.

_____

_____

_____

(2 marks)

c. What does the slope of the graph plotted in **2 a** show?

_____

_____

(1 mark)

d. Using the graph plotted in **2 a** determine the value of the spray race by the end of 3½ years.

(1 mark)

e. If the spray race's life span was 6 years, calculate the salvage value of the spray race.

(3 marks)

Continued/…

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SECTION B (20 MARKS)

3. You are provided with a mixture of two varieties of maize seeds in a container.

a. What marketing function can be performed to show that there are two varieties of maize seeds in the container.

_____

(1 mark)

b. Perform the marketing function in **2 a.**

Draw your results below:

(2 marks)

c. Explain any **one** advantage that farmers will find when they sell the maize after performing the marketing function in question **2 b.**

_____

_____

_____

(2 marks)

d. Using a stone crush the maize seeds after performing the marketing function , observe and record the results.

_____

_____

_____

(2 marks)

Continued/...

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3. (Continued)

e. Which seeds can be recommended for growing in an area where weevils are common? Give a reason for your recommendation.

Recommendation: _____

_____ **(1 mark)**

Reason: _____

_____ **(2 marks)**

4. You are provided with the following specimens:

- a stem cutting used for propagating crops labelled **X**.
- two soil samples labelled **Y** and **Z**.

a. Prepare a planting material from the specimen **X**.

(i) Draw the prepared planting material.

(3 marks)

(ii) On the diagram, label **two** parts that will show that the specimen has been prepared as a planting material. **(2 marks)**

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4. **(Continued)**

b. Rub each of the soil samples between the thumb and forefinger.

(i) Write your observation:

_____

_____

_____

(2 marks)

(ii) From your observation, which of the soil samples can be recommended to be used for growing of the planting material?

_____

(1 mark)

(iii) Give a reason for the answer to question 4 b (ii).

_____

_____

_____

(2 marks)

END OF QUESTION PAPER

This paper contains 8 printed pages.

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